

Contexts As Analyzable Artifacts

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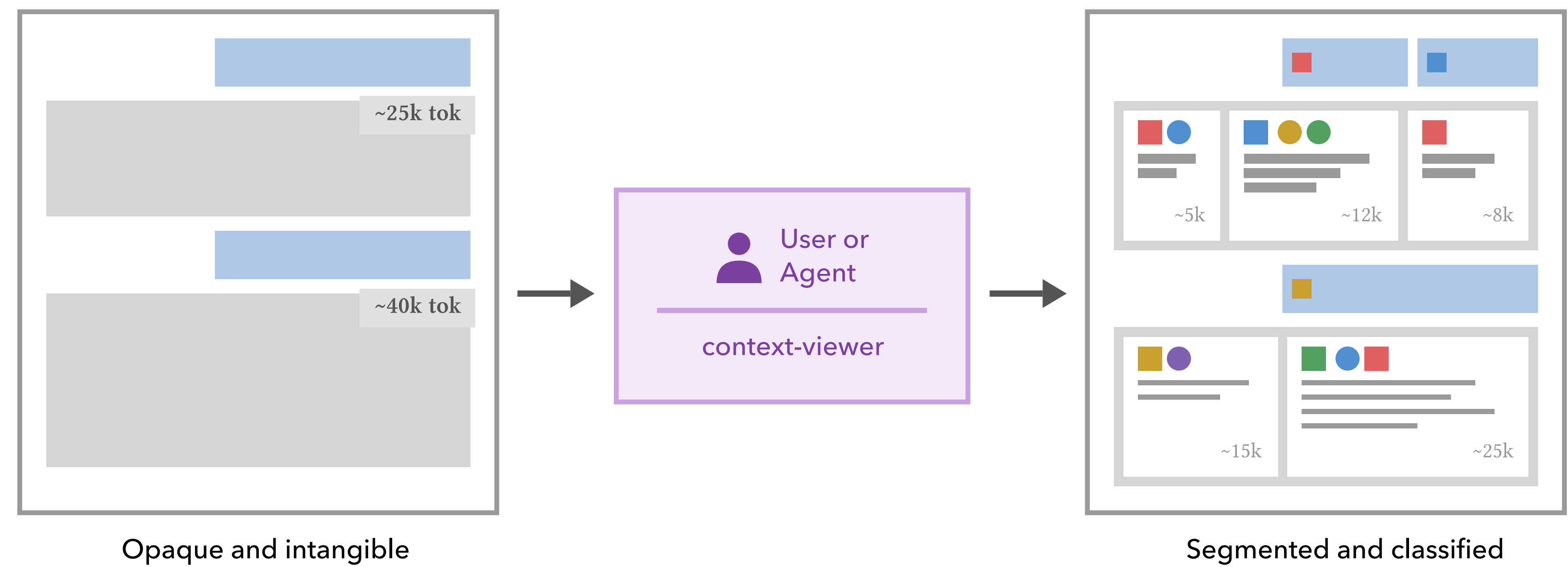
scan for code



What this is

LLM systems depend on long, tool-rich contexts, but those contexts are usually visible only as transcripts or JSON logs.

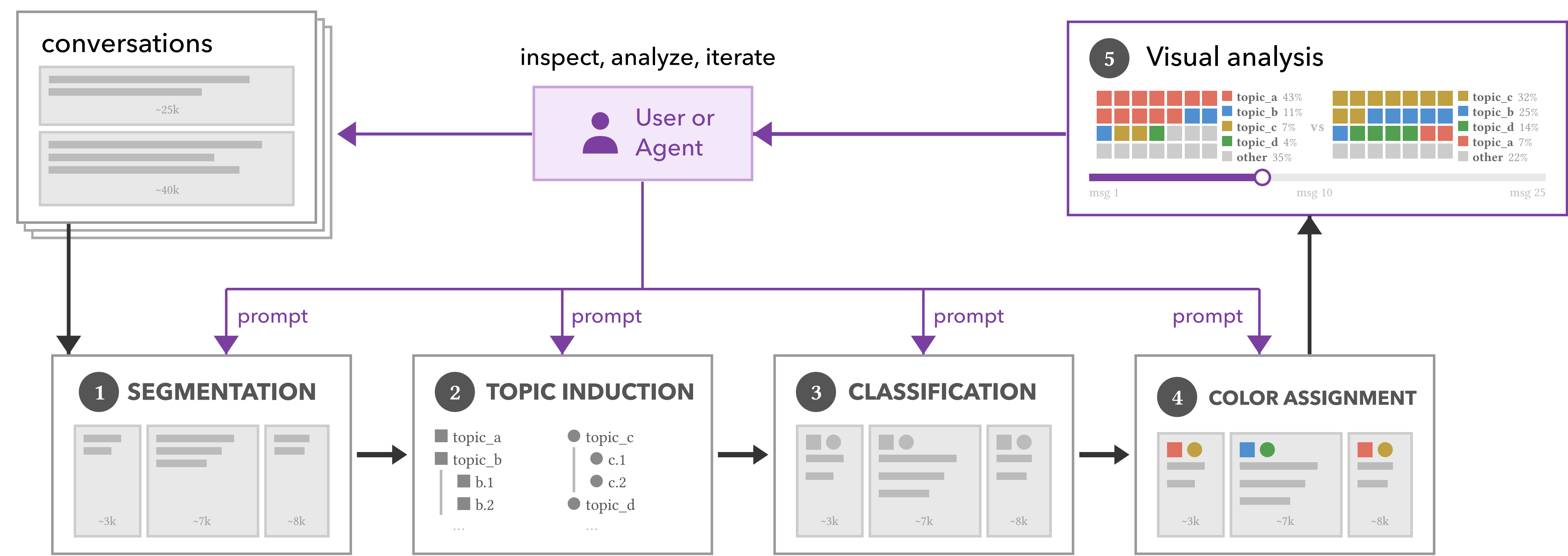
Context Viewer decomposes them into semantic units so developers can inspect where tokens go, what persists, and how trajectories differ.



System Overview

An automated pipeline decomposes context into meaningful constituent components.

A user or an agentic harness can modify this pipeline at every stage to refine the decomposition and steer exploration.

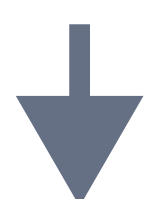
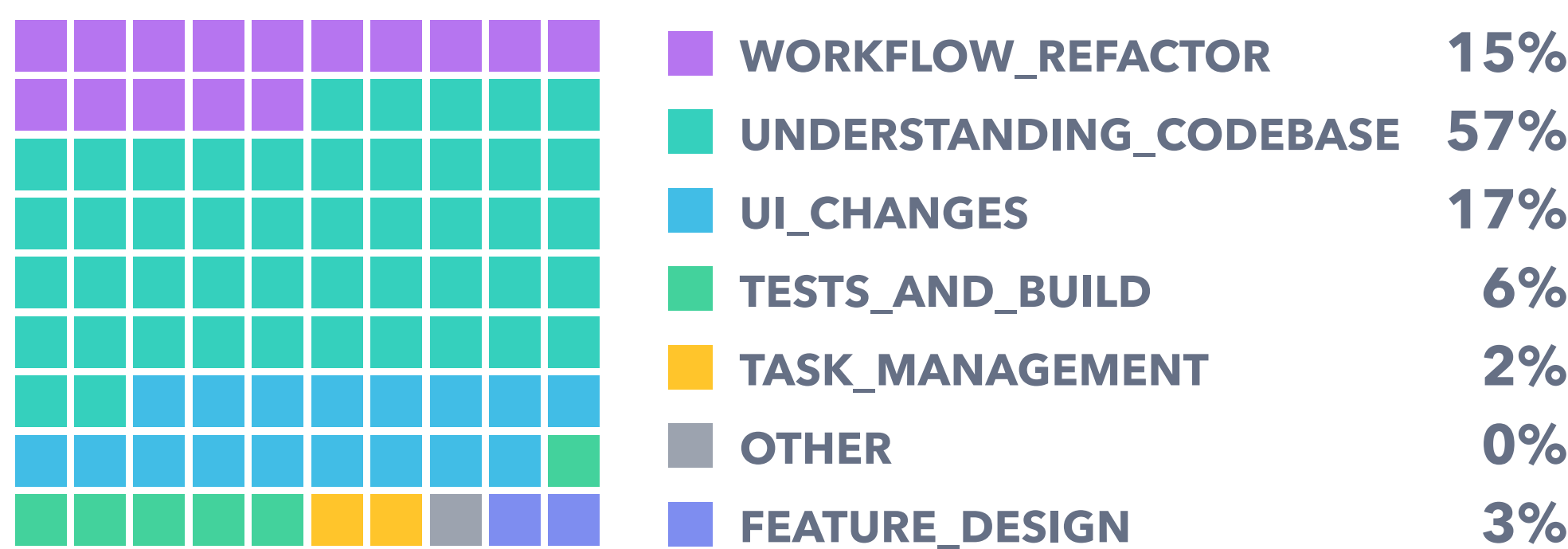


Demo

Analyzing compaction quality in Claude Code

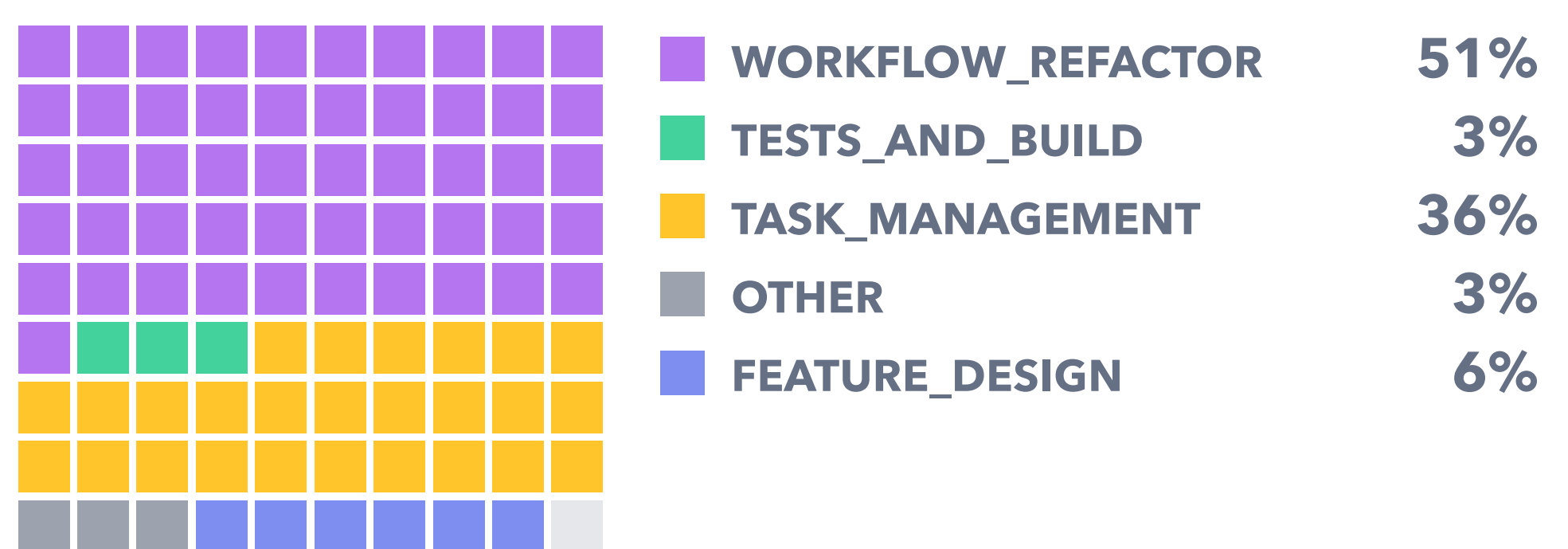
Before Compaction

111,348 TOKENS • 455 MESSAGES



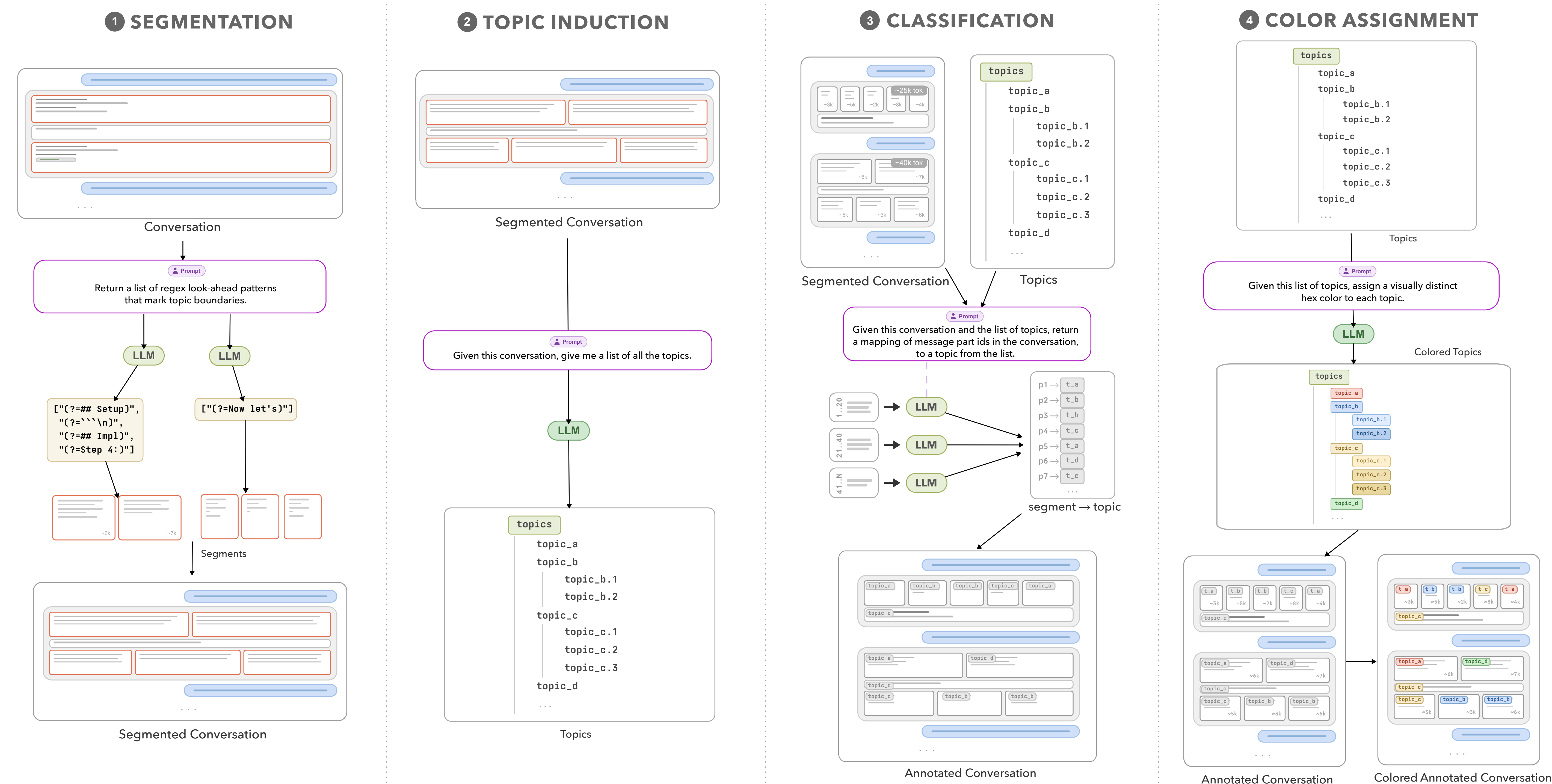
After Compaction

3,122 TOKENS • 1 MESSAGES



The above is an analysis of a compaction event in Claude Code by comparing the token representation of topics in the context before and after compaction. The context before compaction is a 16MB JSON trajectory, while the context after compaction is a 13KB summary in plain text. Context Viewer is able to run the same pipeline across both these contexts and provide a visual comparison that is useful for analysis.

This analysis shows the reduction in composition of codebase understanding, and increase in task management topics.



Contributions

Reframes LLM contexts as an inspectable engineering artifact

Uses LLM-assisted segmentation, topic identification, and classification to analyze context.

Supports practical tasks such as compaction analysis, failure diagnosis, and trajectory comparison.

What we can answer

What exactly survives compaction?

Which failures are caused by the model, and which by the environment?

How does a system prompt or harness evolve over time?